



# **PES UNIVERSITY**

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**Capstone Project Report Phase - 2**

**on**

## **WBPA-QA**

**Submitted by**

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**FACULTY OF ENGINEERING  
DEPARTMENT OF COMPUTER APPLICATIONS  
PROGRAM - MASTER OF COMPUTER APPLICATIONS**

***Certificate***

This is to certify that the project entitled

**WBPA-QA**

is a bonafide work carried out by

**Parth Raheja(PES1PG23CA200)**

in partial fulfillment for the completion of **Capstone Project Phase - 2** work in the Program of Study MCA under rules and regulations of **PES University**, Bengaluru during the period **March 2025 - June 2025**. The project report has been approved as it satisfies the academic requirements of 4<sup>th</sup> semester MCA.

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# Declaration

I, **Parth Raheja**, bearing **PES1PG23CA200** hereby declare that the project entitled, **WBPA-QA**, is an original work done by me under the guidance of **Mr. Dilip Kumar Maripuri**, Associate Professor, PES University and is being submitted in partial fulfillment of the requirements for completion of 4<sup>th</sup> Semester course work in the Program of Study MCA.

All corrections/suggestions indicated for internal assessment have been incorporated in the report.

The plagiarism check has been done for the report and is below the given threshold. I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other course.

Place: Bengaluru

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# Abstract

Water pollution is a growing concern, especially in urban areas like Bangalore, where lake contamination significantly impacts public health and ecosystems. As rapid urbanization and industrialization continue, the quality of water bodies deteriorates due to unchecked pollution levels. Monitoring water quality efficiently is crucial for ensuring safe water consumption and ecosystem sustainability. Traditional water assessment methods are often time-consuming and lack predictive insights, making it essential to develop a more efficient and scalable solution.

This project introduces WBPA-QA (Water Body Properties Analyzer for Quality Assessment), an intelligent system that utilizes machine learning to predict water quality and assess potential health risks. The system integrates historical datasets and user-provided inputs to generate accurate predictions of key parameters such as pH, Total Dissolved Solids (TDS), and Use-Based Class. It is developed using a robust technology stack, including Express.js for the backend, Flask for the machine learning model, React.js for the web application. MongoDB serves as the primary database, ensuring structured storage of processed water quality data. The system's user interface is designed to provide seamless interaction, enabling both general users and authorities to analyze water quality effortlessly.

WBPA-QA successfully delivers high-accuracy predictions, classifies water quality based on standard indices, and offers actionable recommendations to authorities and the public. The application empowers users to make informed decisions regarding water consumption and pollution control. Future enhancements will focus on integrating real-time IoT sensors for continuous data collection and leveraging adaptive AI models to further improve accuracy and reliability. With its scalable architecture and predictive capabilities, WBPA-QA stands as a promising solution to address the growing challenges of water quality management.

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WBPA-QA

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# Chapter 1: Introduction

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# Chapter 1

## Introduction

### 1.1 Problem Scenario

Especially in fast urbanising areas like Bangalore, where industrial discharge, sewage contamination, and urban runoff have greatly polluted water bodies, water pollution is a major environmental problem. Public health, aquatic ecosystems, and general environmental sustainability all depend critically on the declining water quality. Growing worries notwithstanding, current water monitoring systems are still small in scope and sometimes rely on ineffective, time-consuming manual sample methods lacking real-time data. Moreover, the absence of predictive analytics delays early intervention, which makes it challenging for authorities and citizens to act quickly to lower water pollution hazards.

### 1.2 Proposed Solution

Using machine learning algorithms to examine past and user-provided data, WBPA-QA (Water Body Properties Analyser for Quality Assessment) is an advanced water quality prediction system. Providing timely and useful insights, this system is meant to forecast important water quality indicators including pH, Total Dissolved Solids (TDS), and Use- Based Class. React.js for the web interface, React Native for mobile accessibility, Express.js for the backend, and Flask to run the machine learning-driven predictions are among the latest technologies stacked here for the application. Unlike traditional water monitoring systems, WBPA-QA guarantees accurate, real-time information about water quality trends by combining predictive analytics and tailored health recommendations, therefore assuring government authorities and the general public with correct, real-time knowledge about water quality trends.

## 1.3 Purpose

WBPA-QA's main objective is to close the discrepancy between public awareness and water quality evaluation by means of an artificial intelligence-driven system with real-time forecasts. This project helps authorities to make wise judgements on water management and conservation initiatives by means of data-driven insights and trend analysis of water quality. Customised health advice based on water quality assessments also assist individual consumers take required care to guarantee safe water use. From a Computer Science standpoint, WBPA-QA shows how predictive models could support public health and ecological sustainability as well as the pragmatic implementation of machine learning in environmental monitoring.

## 1.4 Scope

Using a mix of historical data and real-time user inputs, the initiative is concentrated in tracking and forecasting water quality for important water bodies in Bangalore. Two main user groups are intended for the system: general users looking for knowledge on the safety of their local water sources and government bodies in charge of water quality control. To ascertain if the water is fit for drinking or causes health hazards, the prediction model assesses important factors. Certain restrictions do exist, though, such reliance on current data sources and the possible requirement for IoT-based sensor integration in further developments to expand real-time monitoring capacity.

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WBPA-QA

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## Chapter 2: Literature Survey

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# Chapter 2

## Literature Survey

### 2.1 Domain Survey

#### 2.1.1 Introduction to the Domain

Public health and environmental sustainability depend on water quality monitoring in great part. Rising urbanisation, industrialisation, and climate change are degrading the quality of surface and groundwater resources at a fast pace. Manual sampling and laboratory testing are time-consuming, costly, and devoid of real-time insights traditional approaches of water quality monitoring entail.

Particularly machine learning and artificial intelligence, recent developments in data-driven technologies have made scalable and more effective solutions for water quality prediction and monitoring possible. Predictive models can examine important characteristics such pH, Total Dissolved Solids (TDS), and turbidity by using historical data and real-time inputs, therefore enabling authorities and the public to make informed judgements.

#### 2.1.2 Core Terminologies and Definitions

Developing a good assessment system depends on an awareness of the basic ideas about predictive analytics and water quality. Here are some important terminologies:

- **Total Dissolved Solids (TDS):** Measures the concentration of dissolved substances in water, affecting its usability.
- **pH Level:** Indicates water's acidity or alkalinity, therefore affecting aquatic life and usage.
- **Use-Based Class:** Based on its fit for drinking, irrigation, and industrial uses, water quality.



- **Machine Learning (ML):** Machine learning (ML) makes predictions and identifies trends by means of algorithms learning from data.
- **Waterborne Diseases:** Included among the diseases transmitted by polluted water are typhoid and cholera.

### 2.1.3 Associated Theoretical Concepts

Several machine learning and data science approaches are used in water quality prediction to examine and interpret challenging environmental datasets. Correct forecasts and significant insights depend critically on several theoretical ideas. These cover geospatial analysis for spatial mapping, time-series analysis for trend forecasting, feature engineering for parameter selection, and supervised learning models for categorisation.

- **Supervised Learning:** Machine learning methods such Random Forest and Support Vector Machines (SVM) are extensively applied for classifying water quality depending on previous data trends.
- **Time-Series Analysis:** Time-series analysis is a method for predicting future fluctuations in parameters including pH, turbidity, and Total Dissolved Solids (TDS) by means of previous water quality trends.
- **Feature Engineering:** Choosing the most important water quality characteristics (pH, TDS, conductivity) reduces dataset noise, hence enhancing forecast accuracy.
- **Geospatial Analysis:** This approach visualises and maps water quality changes over many sites, therefore helping authorities to more precisely identify pollution sources.

### 2.1.4 Challenges and Open Problems

Water quality evaluation presents various difficulties even with developments in IoT-based monitoring systems and machine learning. These difficulties affect the scalability, dependability, and correctness of current prediction models.

- **Lack of Real-Time Data:** Many of the current water quality databases are historical, hence without more IoT sensor integration real-time prediction is challenging.
- **Variability in Data Quality:** Variations in water sampling methods and sensor accuracy provide disparities that influence model performance in data quality.

- **Interpretability of ML Models:** Complex machine learning models operate as black-box systems, which makes it difficult for legislators and scholars to grasp their decision-making procedures.
- **Integration with IoT Sensors:** A strong water quality prediction system depends on real-time monitoring using IoT sensors, which is typically costly and challenging to implement widely.

## 2.2 Related Work

### 2.2.1 Research Papers Review

1. **R. T. V. Ramachandra, et al., "Monitoring and Analysis of Lakes in Karnataka", Energy and Wetlands Research Group, IISc, 2020.**

Available: IISc

- **Paper Overview**

Driven mostly by fast urban development, deforestation, and poor wastewater treatment systems, this thorough study explores the progressive ecological degradation of urban and peri-urban lakes in Karnataka. It describes how treated home sewage and industrial pollutants cause eutrophication, loss of biodiversity, and freshwater source contamination. The need of building an integrated monitoring system to help to restore and protect aquatic environments is underlined in this research.

- **Key Objectives**

- Finding main causes of human interference degrading lake ecosystems.
- Examining temporal and regional fluctuations in water quality with field data and satellite images.
- Suggesting, grounded on empirical data, sustainable conservation strategies.

- **Proposed Methodology**

The method combines on-ground physicochemical examination with geographical study utilising satellite data. Several lakes' worth of key parameters— Biological Oxygen Demand (BOD), Dissolved Oxygen (DO), pH, Total Dissolved Solids (TDS), and nitrate levels—were measured. To link urban development with environmental degradation, the study also used Land Use Land Cover (LULC) change detection.

- **Main Findings/Results**

Results show that more than 85% of the urban lakes under observation were quite con-

taminated, acting as sewage collecting tanks. Consistent observations were of declining vegetation cover, decreasing water surface area, and increasing nutrient loading. The study revealed a clear correlation between low water quality and invasion.

- **Impact and Relevance**

The need of real-time monitoring and the application of remote sensing in water resource management is underlined by this research. Closely aligned with WBPA-QA, it provides a reference for policy interventions, distributed wastewater treatment, and AI-powered predictive analytics for future lake health assessment.

2. **V. P. Harigovindan, et al., "Water Quality Prediction for Smart Aquaculture Using Hybrid Deep Learning Models", IEEE Transactions on Systems, Man, and Cybernetics, 2021.**

Available: ResearchGate

- **Paper Overview**

This article presents a complex hybrid deep learning system to predict water quality measurements in smart aquaculture systems. Understanding that aquatic life is sensitive to physicochemical changes, the research emphasises the need of predictive systems able to identify early departures from ideal water conditions. It closes the distance between smart automation in aquaculture projects and conventional water quality monitoring.

- **Key Objectives**

- Create a strong artificial intelligence model able to predict water quality developments going forward.
- By means of proactive intervention, lower operating hazards and enhance fish health.
- Show how well spatial and temporal learning mix in sequential sensor data.

- **Proposed Methodology**

For spatial feature extraction, the system combines Long Short-Term Memory (LSTM) networks for time-series dependencies with Convolutional Neural Networks (CNNs). IoT sensors in real-time gather parameters on water quality including temperature, pH, dissolved oxygen, and ammonia levels. Trained and verified on this multivariate dataset, the hybrid CNN-LSTM model has accuracy assessed using RMSE and MAPE measures.

- **Main Findings/Results**

Especially in terms of temporal variations, the model obtained better prediction accuracy than standard ML methods. With real-time deployment showing over 90%

accuracy in forecasting important thresholds, immediate mitigating actions as aeration or feeding modification were made possible.

- **Impact and Relevance**

This study shows how by automating decision-making and enhancing sustainability, AI-driven systems might transform aquaculture. Emphasising the synergy between sensor-based data collecting and deep learning analytics, its modular methodology can be modified for more general environmental monitoring systems including WBPA-QA.

3. **Sathish Pasika, et al., "Smart Water Quality Monitoring System with Cost-effective IoT", Heliyon, vol. 6, no. 8, 2020.**

Available: ResearchGate

- **Paper Overview**

This work presents a reasonably priced IoT-based water quality monitoring system catered for underdeveloped and rural areas. Using low-power sensors and GSM-based connectivity, it solves the high cost of conventional water testing techniques by allowing real-time data collecting and analysis free from need on advanced infrastructure. The study stresses the need of scalable, user-friendly solutions to enable communities in guarantee of safe water access.

- **Key Objectives**

- To provide a low-maintenance, reasonably priced system for ongoing observation of vital water parameters.
- To allow remote access and data visualisation via cloud dashboards and mobile apps.
- To help underprivileged and isolated areas identify contamination early on.

- **Proposed Methodology**

Low-cost sensors built inside the system track pH, turbidity, TDS, and temperature. These sensors send real-time data to a centralised cloud platform via an integration with a microcontroller (NodeMCU) and GSM module. The architecture calls for a data pipeline covering preprocessing, aggregation, and web interface-based visual display. Threshold breaching sets off alerts.

- **Main Findings/Results**

In low-bandwidth environments, the system proved to be quite dependable and efficient in keeping constant functioning with minimum power use. Ideal for ongoing community

surveillance, this technology lowered expenses by more than 70% and allowed early identification of aberrant findings as compared to standard laboratory-based testing.

- **Impact and Relevance**

This paper confirms the possibility of reasonably priced IoT systems in democratising access to monitoring of clean water. Its interaction with machine learning models—like those of the WBPA-QA platform—can improve prediction accuracy, automate alarms, and enable data-driven decision-making in distant locations. It provides a robust starting point for applying distributed water quality intelligence.

4. **Sumalatha, et al., "A Study on Contamination of Ground and Surface Water Bodies by Leachate Leakage from a Landfill in Bangalore", Geo-Engineering, 2018.**

Available: ResearchGate

- **Paper Overview**

The environmental effects of leachate leaking from landfills on surrounding surface and groundwater sources in Bangalore are examined in this work. As rainwater passes through solid garbage, leachate—a hazardous liquid—carries dissolved and suspended pollutants into the water table. The study emphasises the growing worry about improper disposal methods and their consequences for human health as well as the ecological.

- **Key Objectives**

- The degree of leachate percolation-induced water pollution
- List important contaminants and their levels in surface and groundwater bodies
- Also, find relationships between landfill vicinity and pollution level.

- **Proposed Methodology**

Groundwater and surface water samples were gathered systematically from different distances from the waste site. With an eye towards physicochemical and biological markers including Chemical Oxygen Demand (COD), Biological Oxygen Demand (BOD), heavy metals (including cadmium and lead), total dissolved solids (TDS), and pH, laboratory analysis Spatial distribution was visualised using geographic information system (GIS) mapping.

- **Main Findings/Results**

With values of COD and BOD above allowed limits, results revealed notable deterioration in water quality close to dump sites. Crucially important quantities also revealed

signs of heavy metal elements. The decreasing pollution values with distance confirmed landfill leachate as the main cause. Emphasising seasonal escalation during monsoon seasons, the study

- **Impact and Relevance**

The results underline how urgently designed landfills with appropriate lining and leachate collecting systems are needed. This work provides basic baseline data for prediction systems such as WBPA-QA, where seasonality, landfill proximity, and groundwater depth allow machine learning models to forecast zones of contamination risk.

5. **Ravikumar P., et al., "Water Quality Index to Determine the Surface Water Quality of Sankey Tank and Mallathahalli Lake, Bangalore", *Applied Water Science*, vol. 7, pp. 1345–1355**

Available: Springer

- **Paper Overview**

The surface water quality of Sankey Tank and Mallathahalli Lake in Bangalore is investigated in this study article with the Water Quality Index (WQI) method. The study underlines the rising worries about urban lake contamination and stresses the use of WQI as a composite indicator to assess the fit of water for home, leisure, and ecological needs. To speed assessments and comparisons, it combines multi-parameter testing into a condensed index.

- **Key Objectives**

- Using WQI approach, ascertain the general water quality condition of two main lakes located in Bangalore.
- Look into seasonal fluctuations and trends in important water quality criteria.
- Sort water usefulness according to conventional classification—Excellent, Good, Poor, etc.).

- **Proposed Methodology**

Pre-monsoon, monsoon, and post-monsoon seasons saw water samples taken from several places around both lakes. In the lab, we analysed parameters including pH, turbidity, dissolved oxygen (DO), total hardness, nitrates, and BOD. Using the weighted arithmetic mean approach, the WQI was computed assigning relative weights to every parameter depending on its health impact.

- **Main Findings/Results**

According to the WQI ratings, Mallathahalli Lake and Sankey Tank both fall into the

”Moderately Polluted” category. Surface runoff bringing urban waste and sewage into the lakes during the monsoon season elevated pollution levels. Although partial maintenance efforts helped Sankey Tank display somewhat better water quality, Mallathahalli Lake revealed poor dissolved oxygen levels and high nitrate concentrations.

- **Impact and Relevance**

The study shows how seasonal dynamics affect pollution level and provide a necessary baseline for lake water quality in urban environments. Artificial intelligence-driven solutions like WBPA-QA for ongoing water quality assessment can incorporate its methodical WQI paradigm. The results also support wise policymaking and prioritising of lake restoration projects all around Bengaluru.

6. **Sudhakar M. Rao, et al., ”Examining Bengaluru’s Potable Water Quality”, Indian Institute of Science Journal, 2024.**

Available: ResearchGate

- **Paper Overview**

With an eye towards WHO and BIS (Bureau of Indian Standards) compliance, this study looks at the drinking water quality in several areas in Bengaluru. The report examines differences in water delivery systems around the city and emphasises the public health consequences of drinking contaminated water. It emphasises how urgently monitoring systems and infrastructure enhancement are needed.

- **Key Objectives**

- Evaluate the water’s potability from several parts of Bengaluru.
- Find dangerous elements include chemical residue, heavy metals, and microorganisms.
- Point up areas of great danger and suggest focused intervention plans.

- **Proposed Methodology**

From household and municipal taps spread over numerous administrative areas, water samples were methodically gathered. Physicochemical parameters (pH, turbidity, TDS), microbiological content (colium), and hazardous metals (lead, arsenic, chromium) were measured in laboratories. Results were mapped geographically using GIS and compared with values derived from health-based guidelines.

- **Main Findings/Results**

According to the study, some zones showed good findings while others surpassed allowed

limits for turbidity and bacterial count. Old distribution pipelines and peri-urban regions showed elevated levels of lead and coliums. Seasonal fluctuations significantly affected water quality, especially during monsoon when cross-contamination and leakage raised.

- **Impact and Relevance**

This study offers important new perspectives on the variation of availability to drinkable water in an urban megacity. WBPA-QA's AI technology can include geospatial pollution mapping into predictive alarms in sensitive areas. The results strengthen the case for civic entities investing in water purification infrastructure and real-time water quality monitoring.

7. Eid M.H., et al., "Human Health Risk Assessment Using Monte Carlo Method", **Annual Review of Environment and Resources**, vol. 46, 2021.

Available: ResearchGate

- **Paper Overview**

This work uses the Monte Carlo simulation approach to assess the hazards connected with water pollution on human health. It emphasises the need of evaluating the degree of contamination of drinking water supplies. Incorporating uncertainty and unpredictability improves the accuracy of conventional risk assessment models by which one may better grasp possible long-term health effects.

- **Key Objectives**

- The health hazards resulting from long-term waterborne contamination
- Use probabilistic modelling to reflect exposure level uncertainty.
- Improve public health decision-making using risk estimates grounded in data.

- **Proposed Methodology**

Probabilistic risk evaluations based on pollutant concentrations, intake rates, and exposure time are performed using the Monte Carlo simulation method. Random sampling runs several simulations in order to produce a distribution of risk values. Using sensitivity analysis and cumulative risk models, heavy metals including arsenic and lead as well as microbiological contaminants are assessed.

- **Main Findings/Results**

According to the study, when exposure is long-term and cumulative, even little amounts of pollutants can cause significant health hazards. Comparatively to deterministic models, Monte Carlo-based evaluations were more exact in pointing out worst-case possi-



bilities. The resulting risk distributions are a good tool for guiding the prioritising of mitigating actions.

- **Impact and Relevance**

This work offers a strong basis for including risk analysis into prediction models derived from artificial intelligence such as WBPA-QA. It improves the decision-making ability of smart water quality systems for early interventions and public health warnings and allows real-time health risk grading depending on sensor input.

8. **Shams M.Y., et al., "Water Quality Prediction Using Machine Learning Models", 2023.**

Available: Springer

- **Paper Overview**

The ability of machine learning algorithms to forecast water quality is investigated in this work. Using actual data, it assesses many ML techniques and their accuracy in predicting important water characteristics. The work offers a comparative evaluation of conventional and contemporary methods of environmental modelling.

- **Key Objectives**

- To evaluate the capacity of several ML techniques in forecasting water quality indicators.
- By means of standard metrics like RMSE, MAE, and  $R^2$ .
- To show that these models might be implemented in real-time monitoring systems with viability.

- **Proposed Methodology**

Among the supervised learning methods the researchers applied were neural networks, support vector machines (SVM), and random forest. Training for the models came from past data including pH, turbidity, and dissolved oxygen—physicochemical water characteristics. Cross-validation and test accuracy measures helped to assess performance.

- **Main Findings/Results**

Random Forest and Neural Networks stood out among the tested models since they had strong predicted accuracy and resilience throughout several water conditions. These models generated dependable projections fit for environmental use and efficiently captured non-linear links between parameters.

- **Impact and Relevance**

The work confirms the acceptance of machine learning methods in systems of smart

water quality prediction. It helps to use AI-based solutions for public health advisory systems and proactive environmental monitoring, such as WBPA-QA.

9. Rui Ma, et al., "Methods for Quantifying Interactions Between Groundwater and Surface Water", *Annual Review of Environment and Resources*, vol. 46, 2021.

Available: ResearchGate

- **Paper Overview**

The development of techniques to measure the interactions between surface and groundwater is the main emphasis of this work. Good regulation of pollution and water resources depend on an awareness of these interconnections. The work stresses the requirement of integrated modelling techniques to adequately represent the intricate dynamics of water exchange mechanisms.

- **Key Objectives**

- To enable evidence-based policy design for water resource protection.
- To build accurate and scalable models for predicting pollution transport through linked hydrological systems.
- To identify the processes controlling the exchange between surface and groundwater systems.

- **Proposed Methodology**

The investigators combined geospatial analysis, hydrological modelling, and field sampling. An integrated interaction map was produced by means of tools including remote sensing, groundwater tracers, and flow simulation. Optional inclusion of machine learning methods helped to maximise flow estimates from past data.

- **Main Findings/Results**

Particularly in metropolitan areas with mixed land use, the study revealed that interactions between groundwater-surface water play a crucial part in the dispersion of pollutants. The hybrid approach produced quite accurate forecasts of flow direction and contaminant spread.

- **Impact and Relevance**

The results of this work have applicability for raising water quality prediction system spatial resolution. Future models can improve risk predictions for shared water bodies across urban and rural areas by including this knowledge into systems such as WBPA-QA.

10. **Suman Kumar Sourav, et al., "Urbanization and Seasonal Effects on the Lake Ecosystem Dynamics from 2002 to 2022 in Bengaluru, South India", 2022.**

Available: ScienceDirect

- **Paper Overview**

The effects of urbanisation on Bangalore's lake ecosystems over a 20-year period are investigated in this work. It gauges seasonal variations in biodiversity and water quality. The long-term dataset emphasises important turning points in ecological changes and helps to detect trends of progressive degradation.

- **Key Objectives**

- Analyse how urban growth shapes lake form and water quality over long terms.
- Examine lake ecosystem seasonal fluctuations in physicochemical and biological characteristics.
- Match freshwater body ecological stress indicators with developments in urban development.

- **Proposed Methodology**

The study blends field data with geospatial analysis. Analysed were satellite images from 2002 to 2022 to evaluate shrinkage and intrusion. Seasonal water samples were also gathered from several lakes concurrently and examined for markers like pH, dissolved oxygen, nutrient content, and microbial count.

- **Main Findings/Results**

Especially in peri-urban areas, lake surface area and volume clearly dropped. Because of sewage inflow and runoff, monsoon seasons revealed more pollutant burdens. Urban lakes were clearly less biodiverse than rural ones.

- **Impact and Relevance**

This study offers important proof of how unbridled urban expansion affects freshwater environments. The results confirm the addition of urbanisation elements in AI-based water quality forecasting systems and provide baseline measures for conservation planning.

## 2.2.2 Comparative Study of Research Papers

| Paper                                                                                                                                       | Authors & Year                    | Problem Statement                                                  | Methodology                                  | Key Findings                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------|
| Monitoring and Analysis of Lakes in Karnataka                                                                                               | Ramachandra T.V., et al. (2020)   | Degradation of lakes in Karnataka due of urbanisation              | GIS and remote sensing approaches            | Identified pollution sources impacting lake ecosystems               |
| Water Quality Prediction for Smart Aquaculture Using Hybrid Deep Learning Models                                                            | V. P. Harigovindan, et al. (2023) | Need of precise water quality prediction in aquaculture            | Hybrid Deep Learning Models (CNN + LSTM)     | Reached high accuracy in forecasting important water parameters      |
| Smart Water Quality Monitoring System with Cost-effective IoT                                                                               | Sathish Pasika, et al. (2020)     | High cost of conventional water quality monitoring methods         | IoT-enabled real-time monitoring system      | Improved efficiency in real-time data collecting                     |
| A Study on Contamination of Ground and Surface Water Bodies by Leachate Leakage from a Landfill in Bangalore                                | Sumalatha, et al. (2018)          | Leachate contamination from landfills influencing water quality    | Field sampling and water quality analysis    | Found substantial degrees of pollution in surrounding water supplies |
| Water Quality Index to Determine the Surface Water Quality of Sankey Tank and Malathahalli Lake, Bangalore Urban District, Karnataka, India | Ravikumar P., et al. (2021)       | Need of a systematic index to assess water quality                 | Water Quality Index (WQI) approach           | Classified lakes based on pollution levels                           |
| Examining Bengaluru's Potable Water Quality                                                                                                 | Sudhakar M. Rao, et al. (2022)    | Variability in potable water quality across Bengaluru              | Comparative analysis of water sources        | Identified hazardous drinking water zones                            |
| Human Health Risk Assessment Using Monte Carlo Method                                                                                       | Eid M.H., et al. (2021)           | Effect of water pollution on human health                          | Monte Carlo simulation-based risk assessment | Quantified health risks due to contaminated water                    |
| Water Quality Prediction Using Machine Learning Models                                                                                      | Shams M.Y., et al. (2020)         | Restricted prediction accuracy of traditional water quality models | Machine Learning (Random Forest, SVM)        | Better prediction accuracy for water quality parameters              |
| Methods for Quantifying Interactions Between Groundwater and Surface Water                                                                  | Rui Ma, et al. (2019)             | Challenge in evaluating groundwater-surface water interactions     | Advanced hydrological modeling approaches    | Developed exact quantification methods                               |
| Urbanization and Seasonal Effects on the Lake Ecosystem Dynamics from 2002 to 2022 in Bengaluru, South India                                | Suman Kumar Sourav, et al. (2023) | Impact of urbanization on lake ecosystems over two decades         | Satellite imagery and time-series analysis   | Observed significant degradation due to human activities             |

Table 2.1: Comparison of Research Papers

## 2.2.3 Identified Research Gaps

- **Limited Predictive Capabilities:** Although most research concentrate on real-time monitoring, they lack future trend analysis.
- **Generalization Issues:** Current models do not fit well in new areas and are taught on particular areas.
- **Health Impact Analysis:** Health Impact Analysis: Few models offer direct links between health hazards and water quality.

- **User-Centric Approach:** None of any major system combines predictive powers with public awareness aspects.

## 2.2.4 Comparative Study of Existing Applications

| Application                     | Features                                    | Technology Stack           | Strengths                     | Limitations                                    |
|---------------------------------|---------------------------------------------|----------------------------|-------------------------------|------------------------------------------------|
| BWSSB Real-Time Monitoring      | Sensor-based water quality assessment       | IoT + Web Dashboard        | Provides real-time insights   | Limited to Bangalore region                    |
| WikiWatershed Water Quality App | Community-based data collection             | Mobile Application         | User-friendly and accessible  | Manual data entry affects accuracy             |
| WBPA-QA (Proposed System)       | ML-based predictions + User recommendations | Flask + React.js + MongoDB | Predicts future water quality | Requires IoT integration for real-time updates |

Table 2.2: Comparative Analysis of Water Quality Applications

## 2.2.5 Conclusion

Internet of Things and machine learning have greatly boosted research in water quality prediction. Still lacking, though, are predictive power, generalisation across different areas, and health impact studies. By means of an ML-powered system that not only forecasts water quality but also presents practical advice for authorities and individuals, WBPA-QA seeks to close these gaps. Using real-time data collecting and deep learning helps our system improve decision-making and support environmentally friendly water management.

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## Chapter 3: Hardware and Software Requirements

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# Chapter 3

## Hardware and Software Requirements

### 3.1 Hardware Requirements

The project calls for a system with enough computational capability to effectively support development, testing, and implementation.

#### 3.1.1 Minimum Hardware Specifications

The following table lists the **minimum hardware configuration** required:

| Component         | Specification                 |
|-------------------|-------------------------------|
| Processor         | Intel Core i3 (or equivalent) |
| RAM               | 4GB                           |
| Network Interface | Wi-Fi 802.11ac / Ethernet     |
| Peripherals       | Keyboard, Mouse, Display      |

Table 3.1: Minimum Hardware Requirements

### 3.2 Software Requirements

This project's software stack consists in **development tools, programming languages, and frameworks** guaranteeing seamless implementation and execution.

#### 3.2.1 Development Tools

Following development tools helped to enable effective coding and debugging:

| Software            | Purpose                         |
|---------------------|---------------------------------|
| Visual Studio Code  | Code editor / IDE               |
| Git & GitHub/GitLab | Version control & collaboration |
| Docker (Optional)   | Containerization & Deployment   |

Table 3.2: Development Tools

### 3.2.2 Programming Languages & Frameworks

The project was carried out combining frameworks with several programming languages:

| Technology           | Usage                 |
|----------------------|-----------------------|
| Python 3.8+          | AI models             |
| JavaScript (Node.js) | Server-side scripting |
| Vite + React.js      | Frontend development  |
| Express.js           | Backend development   |
| NoSQL (MongoDB)      | Database management   |

Table 3.3: Programming Languages and Frameworks

### 3.2.3 Additional Libraries & APIs

Integration of several libraries and APIs improved project capability.

| Library/API   | Purpose                 |
|---------------|-------------------------|
| Scikit-learn  | Machine Learning models |
| Pandas, Numpy | Preprocessing and EDA   |
| RESTful APIs  | Backend communication   |

Table 3.4: Additional Libraries and APIs



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# Chapter 4: Software Requirements Specification

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# Chapter 4

## Software Requirements Specification

### 4.1 Users

- **Authorities**

Authorities comprise environmental agencies in charge of tracking water quality as well as governmental entities. They have prediction insights, historical records, and water quality data at hand. They can also examine system generated reports and act appropriately for the control of water quality.

- Track water quality trends at several sites.
- View comprehensive statistics and past performance.
- Create reports on water quality for reference on policies.
- Point up places of great danger for contamination.
- Apply preventive actions grounded on system insights.

- **General Public**

Among general users are those who wish to investigate their surrounds' water quality. Entering their location and surrounding water body information, they can obtain real-time forecasts regarding water quality and any health effects.

- For their location, check expected and real-time water quality.
- Get advice linked to health depending on water quality.
- Report pollution events to authorities.
- Know the patterns in water contamination in their neighbourhood.

## 4.2 Functional Requirements

### FR1. Data Collection and Preprocessing

The system continuously aggregates historical and real-time water quality records from various lakes and water bodies across Bangalore. This data includes essential physicochemical parameters such as pH, Total Dissolved Solids (TDS), Electrical Conductivity, and Turbidity. Prior to analysis, the raw data undergoes extensive preprocessing—removing noise, handling missing values, normalizing values, and structuring datasets for efficient machine learning integration. The preprocessed data is securely stored in a structured database to facilitate retrieval and model training.

### FR2. Water Quality Prediction

Users interact with the system by entering their Station (STN) code, residential location, and the month of interest. Based on this input, the system leverages trained machine learning models to predict key water quality indicators—such as pH and TDS—alongside a Use-Based Water Class (e.g., potable, agricultural, industrial). These predictions empower users with foresight into the potential usability and safety of local water sources.

### FR3. Health Risk Analysis and Recommendations

Following the prediction of water quality, the system assesses associated health risks by mapping contaminant levels against established safety thresholds. It then generates personalized preventive suggestions including changes in dietary habits, usage of specific water filtration systems, and lifestyle practices to reduce vulnerability to waterborne diseases. These recommendations aim to enhance community health and resilience.

### FR4. Interactive Dashboard for Authorities

A web-based dynamic dashboard is made available to municipal authorities, environmental agencies, and policy makers. The dashboard offers access to trends in historical water quality data, real-time status updates across regions, geographic mapping of contamination hotspots, and analytical reports for strategic decision-making. It also includes downloadable summaries and predictive alerts for proactive governance.

**FR5. Web Application for Public Use**

An intuitive web platform is accessible to the general public, allowing citizens to check local water quality reports, receive expert advice based on their location, and file pollution or contamination complaints. The application supports multilingual access, mobile responsiveness, and includes educational content about water safety and conservation.

## 4.3 Non-Functional Requirements

**NFR1. Availability**

The system must be accessible 24/7 to both public users and government stakeholders. High availability is ensured through cloud infrastructure with automated backups, failover mechanisms, and load balancing. Even in the event of partial system outages, critical functionalities like water quality alerts and risk notifications should remain active.

**NFR2. Performance**

The platform should deliver predictions and data insights within 3–5 seconds of user interaction, regardless of dataset size or user load. This includes seamless navigation between dashboards, reports, and input forms on both desktop and mobile devices. Performance optimization techniques such as API caching and lazy loading are employed.

**NFR3. Security and Authentication**

A robust security framework is implemented with JWT-based user authentication to safeguard user sessions. Access to functionalities is controlled via role-based permissions, distinguishing between admin, authority, and general user roles. All data—both personal and environmental—is encrypted during storage and transmission, adhering to data privacy laws and ethical AI guidelines.

**NFR4. Scalability**

The architecture is designed to scale horizontally, allowing seamless expansion

as new regions, datasets, or users are added. It supports containerized deployment on cloud platforms like AWS, Azure, or GCP, and is compatible with future integration of IoT sensors for live water quality monitoring via edge computing.

**NFR5.****Reliability**

System reliability is guaranteed through automated monitoring, anomaly detection, and regular data integrity checks. The application aims for an uptime of at least 99.9%, with automatic logging and alerting of faults or inconsistencies in predictions. Redundant servers and real-time synchronization ensure uninterrupted service delivery.

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# Chapter 5: System Design

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# Chapter 5

## System Design

### 5.1 Architecture Diagram

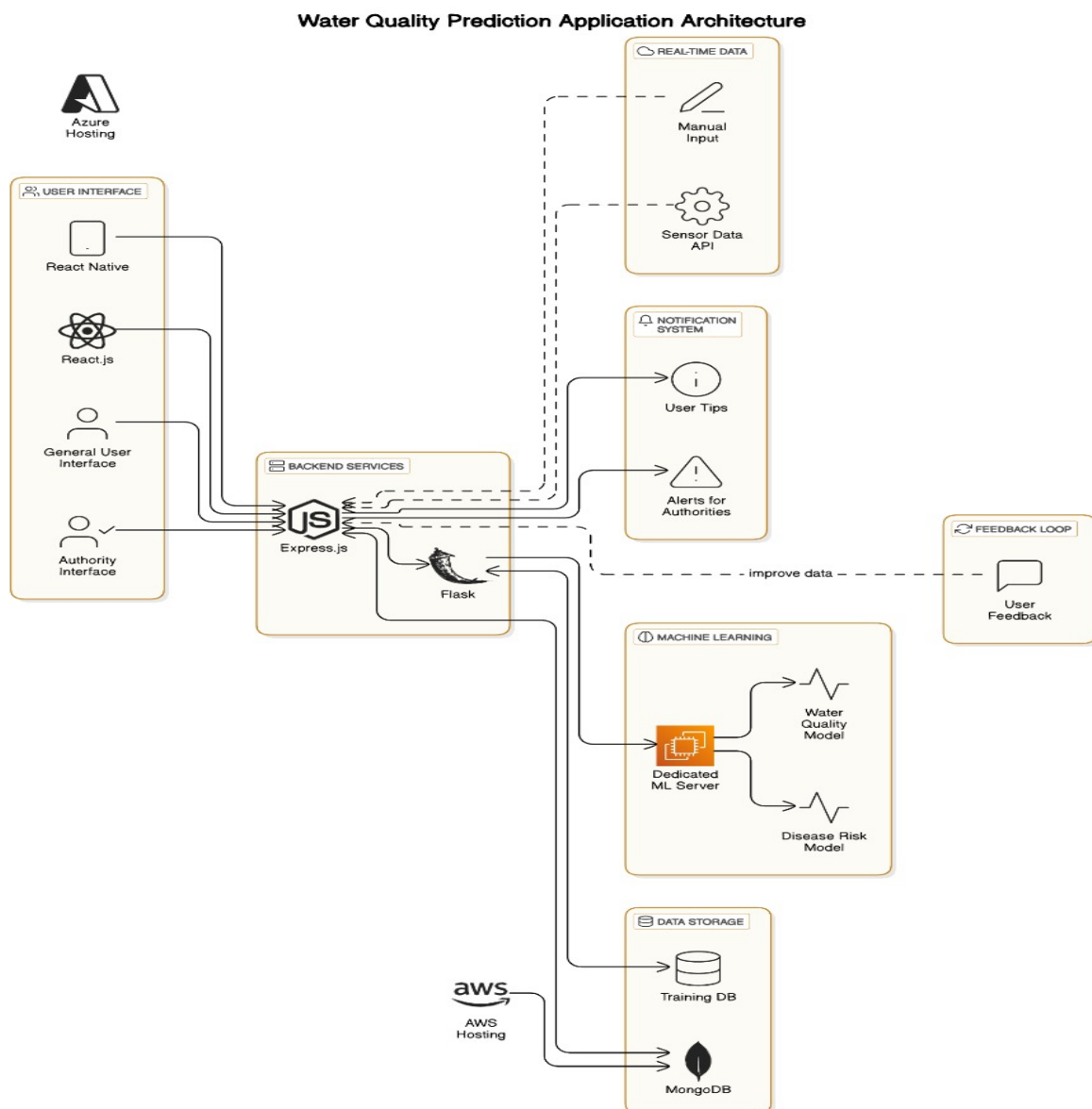


Figure 5.1: Architecture Diagram

The architecture of the Water Quality Prediction Application is organized into four distinct layers, each contributing to the system's overall functionality and scalability:

- **Presentation Layer:**

This layer represents the front-end interface accessible to both the general public and authority users. Built using React.js and React Native, the UI provides a responsive and interactive experience across platforms. General users can report or view water quality data, while authorities are provided with dedicated dashboards to monitor trends and respond to alerts. Features such as 3D visualization, user tips, and feedback forms enhance user engagement and accessibility.

- **Business Logic Layer:**

Acting as the intermediary between the UI and backend services, this layer is implemented using Express.js and Flask frameworks. Express.js handles HTTP requests, session management, and routing of API calls, while Flask is used to interface with the ML server. Core responsibilities include validating user input, authenticating sessions, managing access roles, forwarding sensor and manual data to the ML server, and issuing alerts to the Notification System. It also ensures seamless interaction between data pipelines and storage systems.

- **Machine Learning Layer:**

This layer is hosted on a dedicated ML server, which processes both real-time and historical data to make predictions. It runs two key models: the *Water Quality Model* which assesses potability or contamination risk, and the *Disease Risk Model* which predicts health hazards based on contaminant levels. These models are trained on curated datasets and can adapt using feedback from users and authorities. Predictions from this layer are returned to the business logic layer and then rendered in the UI for end-users.

- **Data Storage Layer:**

Persistent storage is managed using MongoDB, chosen for its scalability and schema-less structure. It stores user inputs, sensor API data, training data for ML models, and feedback logs. The database supports fast retrieval for analytics and model training purposes. AWS hosting provides cloud-based scalability and high availability, ensuring real-time data syncing across modules. The system can store data over long periods to support temporal analysis and improve model accuracy.



## 5.2 Process Flow Diagram

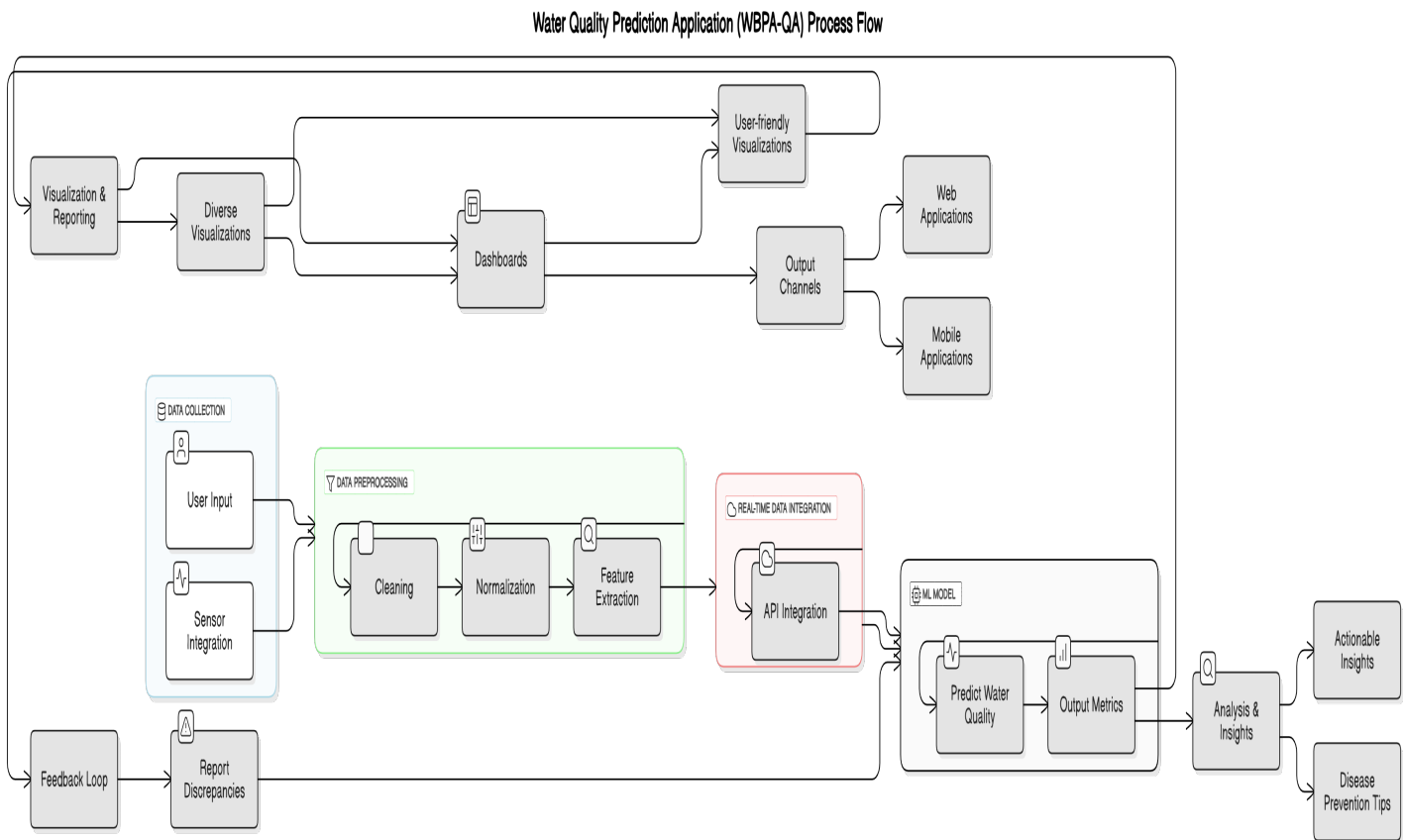


Figure 5.2: Process Flow Diagram

The system flow for the Water Quality Prediction Application is organized into the following key stages:

- **Data Collection:**

The system captures raw inputs through two primary sources: manual user input (location, perceived water condition) and automated sensor integration. This multi-source strategy enhances data accuracy and ensures coverage of both real-time observations and subjective field reports.

- **Data Preprocessing:**

The collected data undergoes preprocessing to improve quality and ensure model readiness. It involves:

- *Cleaning*: Removal of noise, missing values, and invalid entries.
- *Normalization*: Scaling and transformation of data for consistency across features.
- *Feature Extraction*: Derivation of relevant attributes (e.g., turbidity, pH, dissolved oxygen) that influence water quality and disease risk.

- **Real-time Data Integration:**

Preprocessed data is integrated with external sources via API endpoints. This enables synchronization with other platforms or IoT-based water quality sensors, expanding the scope of real-time analytics.

- **ML Model Processing:**

The integrated data is fed into dedicated machine learning models:

- *Predict Water Quality*: Assesses potability, contamination levels, and trends.
- *Output Metrics*: Generates numerical and categorical results for interpretation.

- **Insights and Visualization:**

Model outputs are analyzed for actionable insights. Key results are visualized through interactive dashboards and visual reporting tools, targeting:

- End-users, via mobile and web applications.
- Authorities, via detailed dashboards with drill-down capabilities.

Insights include alerts, health tips, and disease prevention recommendations.

- **Feedback Loop:**

A feedback mechanism allows users and authorities to report discrepancies or corrections. These inputs are recycled into the preprocessing pipeline, enabling continuous improvement and adaptive learning.

This process ensures data integrity, timely analysis, and proactive decision-making to enhance public health response regarding water quality conditions.

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## Chapter 6: Detailed Design

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# Chapter 6

## Detailed Design

### 6.1 Use Case Diagram

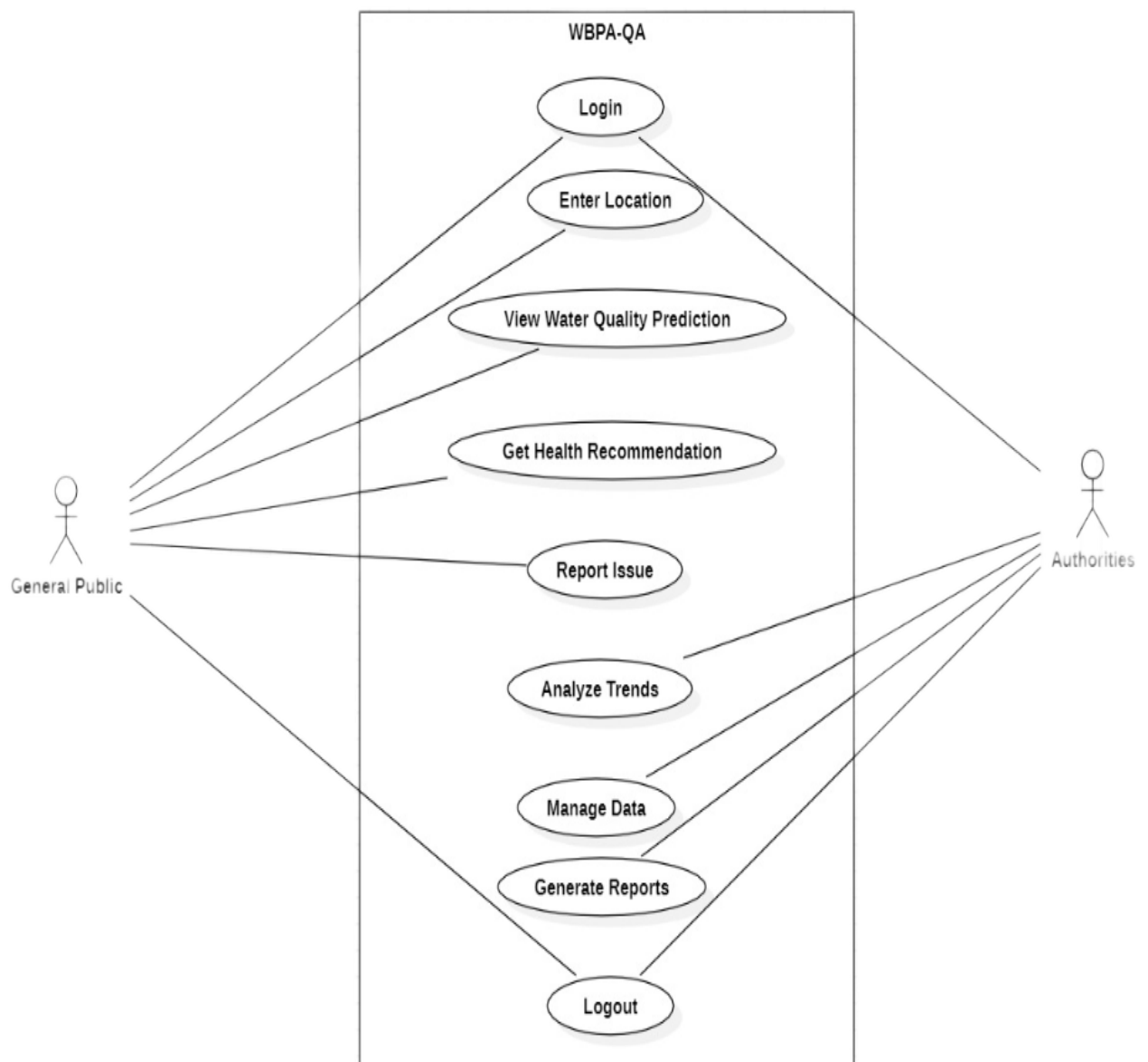


Figure 6.1: Use Case Diagram

The use case diagram illustrates the interactions between the primary actors and the Water Quality Prediction Application (WBPA-QA). The system supports two main categories of users: the **General Public** and **Authorities**. Each actor interacts with a defined set of functionalities, described below:

- **General Public:**

- **Login / Logout:** Users securely sign into the system using personal credentials.
- **Enter Location:** Allows users to input specific geographic details for water quality assessment.
- **View Water Quality Prediction:** Displays results generated by ML models, indicating water safety levels.
- **Get Health Recommendation:** Based on predicted water quality, users receive suggestions related to health risks and precautions.
- **Report Issue:** Users can flag discrepancies or raise concerns about incorrect predictions or water contamination.

- **Authorities:**

- **Analyze Trends:** Allows inspection of historical and real-time data to understand seasonal or location-based patterns.
- **Manage Data:** Enables authorities to upload, edit, and validate datasets coming from different sources.
- **Generate Reports:** Facilitates creation of PDF or dashboard-based reports for decision-making or public communication.
- **Report Issue:** Authorities can also raise flags about system anomalies or critical updates from field visits.

The architecture ensures role-based access, with some overlapping functionalities (such as reporting issues), while safeguarding privileged actions (e.g., data management) exclusively for authorized personnel. This dual-user structure supports both community participation and official oversight.

## 6.2 Sequence Diagram

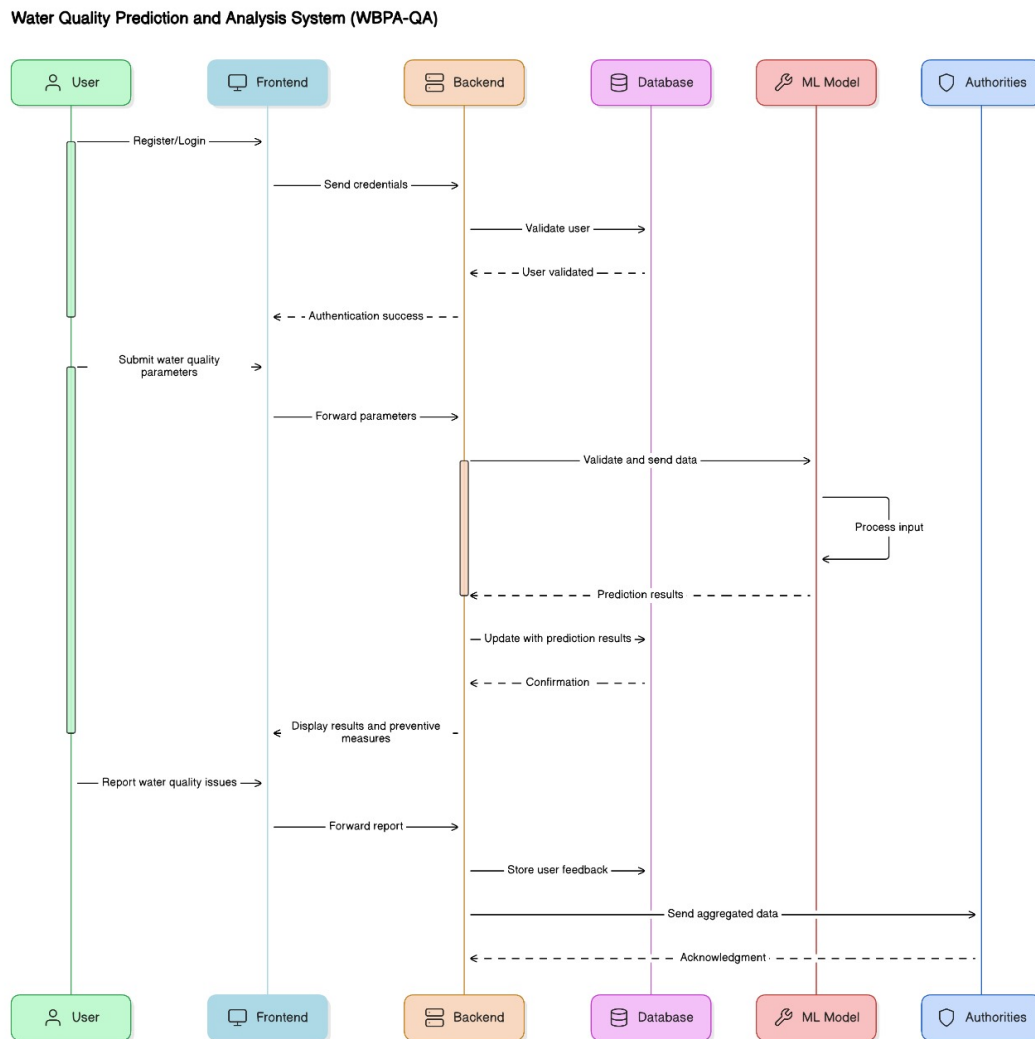


Figure 6.2: Sequence Diagram

The sequence diagram outlines the chronological interaction between the user, frontend, backend, database, machine learning model, and authorities. It provides a clear visualization of how data flows and decisions are executed within the Water Quality Prediction and Analysis System (WBPA-QA). The major steps are as follows:

- **User Authentication:**

The process begins when a user registers or logs in through the frontend interface. The login credentials are securely sent to the backend, which interacts with the database to validate the user. Once authenticated, the user is granted access.

- **Parameter Submission:**

After login, the user submits location details and water quality parameters (such as STN Code or sample characteristics). The frontend captures and forwards these parameters to the backend.

- **Data Validation and Processing:**

The backend validates the parameters and prepares the dataset for analysis. It queries the database for supporting data and sends relevant entries to the ML Model component.

- **Prediction and Result Generation:**

The ML Model receives the inputs, processes the data, and predicts critical indicators such as:

- pH Level
- TDS (Total Dissolved Solids)
- Dissolved Oxygen (O<sub>2</sub>)

These predicted results are returned to the backend.

- **Display of Results and Recommendations:**

Once the results are received, they are updated in the database and shown to the user via the frontend. Health-based recommendations and preventive tips are also displayed based on these values.

- **Issue Reporting and Feedback Loop:**

If discrepancies are noted, the user can report issues through the interface. These reports are forwarded by the backend to be stored in the database. The system ensures a feedback loop to continuously improve model accuracy.

- **Authority Notification:**

Periodically, the backend aggregates user reports and system results. This aggregated data is sent to the authority interface for further analysis and policy response. The authorities acknowledge the receipt and may act upon the insights.

This sequence ensures a secure, responsive, and feedback-driven prediction pipeline that supports both individual health awareness and public health management.

## 6.3 User and Authorities Schema

```

_id: ObjectId('67b2abab73c4db011b828bdf')
email: "parthraheja1205@gmail.com"
password: "$2a$10$zFMT9dIDdXVZ.PsmEvUqe.CfXRYfCX8nUG7b10IgRQwBmWWkjNRM6"
name: "Parth Raheja"
isVerified: true
lastLogin: 2025-02-17T03:23:53.761+00:00
createdAt: 2025-02-17T03:23:23.613+00:00
updatedAt: 2025-02-17T03:23:53.762+00:00
__v: 0
address: "PESU Boys Hostel, BSK 3rd Stage, Bangalore"

```

Figure 6.3: User Schema

### 6.3.1 User Schema

| Field                     | Description                            |
|---------------------------|----------------------------------------|
| Name (String, Required)   | Full name of the user.                 |
| Email (String, Unique)    | Used for authentication.               |
| Password (String, Hashed) | Secured user password.                 |
| Address (String)          | Helps fetch nearby water quality data. |

Table 6.1: User Schema

The User Schema allows authentication and tailored water quality recommendations by storing the key information of average consumers. The system stores passwords in a hashed form, therefore guaranteeing user data security.



### 6.3.2 Authority Schema

| Field                                   | Description               |
|-----------------------------------------|---------------------------|
| Organization Name<br>(String, Required) | Name of the authority.    |
| Address (String, Required)              | Location of the office.   |
| Email (String, Unique)                  | Used for authentication.  |
| Password (String, Hashed)               | Secured authority access. |

Table 6.2: Authorities Schema

Designed for government agencies or organisations in charge of observing water quality is the Authority Schema. It offers access to analytics and reports for water quality evaluation and policy-making and helps to facilitate safe authentication.

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## Chapter 7: Implementation

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# Chapter 7

## Implementation

### 7.1 Pseudocode

#### 7.1.1 Location and Lake Selection

---

**Algorithm 1** Location and Lake Selection

---

**Require:** Predefined mapping of locations to their corresponding lakes, user interface elements (dropdowns)

**Ensure:** Display a filtered list of nearby lakes based on user-selected location

```
1: Initialize an array of locations  $L = [L_1, L_2, \dots, L_n]$ 
2: Initialize a dictionary  $M$  where  $M[L_i]$  returns a list of nearby lakes
3: Render a dropdown UI element with all location options from  $L$ 
4: Disable lake dropdown until location is selected

5: while user has not selected a location do
6:     Wait for user input
7: end while

8: if selected location  $\in L$  then
9:     Fetch corresponding list of lakes from  $M[\text{selected location}]$ 
10:    if lake list is not empty then
11:        Enable lake dropdown
12:        Populate lake dropdown with  $M[\text{selected location}]$ 
13:    else
14:        Show alert: "No lakes mapped to selected location."
15:        Disable lake dropdown
16:    end if
17: else
18:     Show error message: "Invalid location selected."
19:     Reset selection and prompt user again
20: end if

21: On lake selection, store selected lake for future processing

22: Return: Validated user selection for location and lake
```

---

## 7.1.2 Water Quality Data Generation

---

**Algorithm 2** Model-Based Water Quality Prediction

---

**Require:** Validated user inputs: `Location`, `Lake`

**Ensure:** Predict pH, TDS, and Dissolved Oxygen (DO) values using a trained machine learning model

```
1: if Location ≠ NULL and Lake ≠ NULL then
2:   Step 1: Feature Construction
3:   Construct an input feature vector X using contextual data from the selected Location and
   Lake, including geospatial coordinates and historical references (if available)
4:   Step 2: Load Model
5:   Load the pre-trained water quality prediction model MLModel from disk or cache, ensuring
   it has been validated and optimized for production use
6:   Step 3: Predict Parameters
7:   Pass the input vector X to the model: MLModel.predict(X) to obtain predicted values for:
      • pH (acidity/alkalinity level)
      • TDS (Total Dissolved Solids in mg/L)
      • DO (Dissolved Oxygen in mg/L)
8:   Store these predicted values in the object: WaterQualityData = {pH, TDS, DO}
9:   Step 4: UI Integration and Visualization
10:  Generate interactive bar charts or meters based on predicted values for:
      • pH
      • TDS
      • DO
11:  Display the visual results in a responsive layout optimized for both web and mobile screens
12: else
13:   Show alert message: "Please select a valid location and lake to get predictions."
14:   Abort the prediction routine
15: end if

16: Return: WaterQualityData object containing the predicted pH, TDS, and DO values
```

---

### 7.1.3 Disease Risk Prediction via API

---

**Algorithm 3** Disease Risk Prediction via API

---

**Require:** WaterQualityData object (pH, TDS, DO), trained Machine Learning model, Flask-based REST API

**Ensure:** Generate waterborne disease predictions and provide preventive health tips

1: **Step 1: Preprocess Input**

2: Accept validated values for:

- pH (Acidity/Alkalinity of water)
- TDS (Total Dissolved Solids)
- DO (Dissolved Oxygen)

3: Normalize and format data into a structured JSON array:

- { "pH": 7.4, "TDS": 320, "DO": 6.8 }

4: **Step 2: API Request**

5: Send HTTP **POST** request to:

- /api/predict-disease

6: Attach payload as JSON body containing formatted water quality values

7: **Step 3: Model Inference**

8: Flask API receives the request and performs:

1. Deserialize request body into model input
2. Predict categorical label representing water quality risk

9: **Step 4: Disease Mapping**

10: Based on predicted risk class, map to:

- List of likely waterborne diseases (e.g., Cholera, Diarrhea, Typhoid)

11: Retrieve disease-specific preventive tips from local database

12: **Step 5: Display Results**

13: On the frontend:

- Show disease names in bullet list or card view
- Display tips for each predicted disease
- Allow report download as PDF (optional)

14: **Return:**

- Disease prediction list and corresponding health safety recommendations
-

### 7.1.4 Historical Trends Tracking

---

**Algorithm 4** Historical Trends Tracking

---

**Require:** New water quality entry (with pH, TDS, DO values)

**Ensure:** Maintain and visualize the latest 5 entries for historical trend monitoring

1: **Step 1: Retrieve Current State**

2: Fetch existing array of water quality records from:

- Application state (e.g., React useState or Vue reactive state)
- Optional: Retrieve persisted state from browser local storage

3: **Step 2: Validate History Length**

4: **if** current trend array length > 4 **then**

5:     Remove the oldest (first) entry to maintain a rolling window of size 5

6: **end if**

7: **Step 3: Generate New Entry**

8: Create a structured object with:

- Timestamp (e.g., `Date.now()` or `new Date().toLocaleTimeString()`)
- pH, TDS, DO values from current session

9: **Step 4: Update Trend Array**

10: Append the new entry to the trend array

11: **Step 5: Sync to Frontend Chart**

12: Render updated array in historical chart using:

- LineChart component (e.g., Recharts or Chart.js)
- Axes: Time vs. Metric values (pH, TDS, DO)

13: **Step 6: Persist (Optional)**

14: Save updated trend array to:

- Browser local storage or IndexedDB (for offline access and continuity)

15: **Return:** Updated water quality trend array and real-time visualization

---

## 7.1.5 PDF Report Generation

---

**Algorithm 5** Health Report PDF Generation

---

**Require:** User inputs: Location, Lake, Water Quality Metrics (pH, TDS, DO), Disease Predictions

1: **Step 1: Initialize PDF Document**

2: Create new PDF instance using jsPDF or a similar library

3: Set metadata: Title = “WBPA-QA Health Report”, Author = “WBPA-QA System”

4: Add header: Title text + app logo (if available)

5: **Step 2: Add Section - Location and Lake Details**

6: Insert formatted text block:

- **Location:** User-selected region (e.g., Koramangala)
- **Nearby Water Body:** Selected lake or reservoir

7: **Step 3: Insert Water Quality Table**

8: Create a table with 3 rows and 2 columns:

- **Parameters:** pH, TDS (mg/L), Dissolved Oxygen (mg/L)
- **Values:** Average of the 3 generated samples

9: Apply light gray borders and shaded header row for style

10: **Step 4: Add Section - Disease Predictions**

11: Write section heading: “Potential Waterborne Diseases and Precautions”

12: **for** each disease in prediction list **do**

13:     Print disease name in bold

14:     Include 2–3 lines of description

15:     Add bullet list for Preventive measures.

16: **end for**

17: **Step 5: Apply Styling**

18: Use professional fonts: Helvetica or Courier

19: Set title and section headers to WBPA-QA theme blue

20: Align text blocks with left margin; apply consistent padding

21: Use appropriate spacing to avoid content overflow

22: **Step 6: Export and Download**

23: Finalize page layout

24: Trigger file download: `wbpa.health.report.pdf`

25: **Return:** Stylized multi-section PDF containing user location, lake, water metrics, disease risks, and preventive measures

---

## 7.2 Code Logic

The following code demonstrates the core logic of the implementation.

### 7.2.1 Python Code for Machine Learning Model

```

1 # Constants
2 CSV_FILE = "water_quality.csv"
3 MODEL_FILE = "water_quality_model.pkl"
4 TARGET_COLUMNS = ["Total Dissolved Solids (mg/L)", "pH", "Use Based Class"] #
    Update as per your CSV
5
6 def sanitize_columns(df):
7     """
8     Clean column names by removing unwanted characters and spaces.
9     """
10    df.columns = df.columns.str.replace(r'\s+', ' ', regex=True)
11    df.columns = df.columns.str.replace(r'\r\n|\r|\n', '', regex=True)
12    df.columns = df.columns.str.replace(r'[^w\s]', '', regex=True)
13    return df
14
15 def preprocess_data(df):
16     """
17     Preprocess data: encode categoricals and ensure numerics.
18     """
19     if "Month" in df.columns:
20         df["Month"] = df["Month"].astype("category").cat.codes
21     if "Use Based Class" in df.columns:
22         df["Use Based Class"] = df["Use Based Class"].astype("category").cat.
codes
23     df = df.apply(pd.to_numeric, errors="coerce").fillna(0)
24     return df
25
26 def train_model_from_csv():
27     # Load and sanitize dataset
28     df = pd.read_csv(CSV_FILE)
29     df = sanitize_columns(df)
30     df = preprocess_data(df)
31
32     print("Columns in the dataset:", df.columns.tolist())
33     print("Sample data:")
34     print(df.head())
35
36     # Validate target columns
37     missing = [col for col in TARGET_COLUMNS if col not in df.columns]
38     if missing:
39         print(f"Missing columns in CSV: {missing}")
40         return
41
42     # Feature & target separation
43     X = df.drop(columns=TARGET_COLUMNS, errors="ignore")
44     y = df[TARGET_COLUMNS]
45
46     # Train-test split
47     X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
48
49     # Train regressor
50     model = RandomForestRegressor(n_estimators=100, random_state=42)

```



```
51     model.fit(X_train, y_train)
52
53     # Evaluate
54     y_pred = model.predict(X_test)
55     mse = mean_squared_error(y_test, y_pred, multioutput="raw_values")
56     print("Mean Squared Error for each target:", mse)
57
58     # Save model
59     joblib.dump(model, MODEL_FILE)
60     print(f"Model saved to {MODEL_FILE}")
61
62 if __name__ == "__main__":
63     train_model_from_csv()
```

Listing 7.1: ML Model for Water Quality Prediction

## 7.2.2 Python Code for Disease Risk Prediction API

```
1
2 app = Flask(__name__)
3 model = joblib.load("water_quality_model.pkl")
4
5 @app.route('/predict', methods=['POST'])
6 def predict():
7     data = request.get_json()
8     features = [data['pH'], data['TDS'], data['DO']]
9     prediction = model.predict([features])
10    return jsonify({'predicted_quality': prediction[0]})
11
12 if __name__ == '__main__':
13     app.run(debug=True)
```

Listing 7.2: Flask API for Disease Prediction

## 7.2.3 JavaScript Code for Web Dashboard Logic

```
1 async function fetchWaterQuality(location, lake) {
2     const response = await fetch('/api/water-quality', {
3         method: 'POST',
4         headers: { 'Content-Type': 'application/json' },
5         body: JSON.stringify({ location, lake })
6     });
7
8     const data = await response.json();
9     updateCharts(data.metrics);
10    displayHealthRisks(data.diseases);
11 }
12
13 function updateCharts(metrics) {
14     // Use Chart.js or Recharts to visualize pH, TDS, DO
15     drawBarChart(metrics.pH, "pH Levels");
16     drawBarChart(metrics.TDS, "TDS Levels");
17     drawBarChart(metrics.DO, "Dissolved Oxygen");
18 }
```

Listing 7.3: Fetch and Display Water Quality Data

## 7.2.4 JavaScript Code to Download PDF Report

```

1 function downloadReport(report) {
2   const doc = new jsPDF();
3   doc.setFont("helvetica", "bold");
4   doc.text("Water Quality Health Report", 14, 20);
5
6   autoTable(doc, {
7     startY: 30,
8     head: [["Location", "Lake", "Quality"]],
9     body: [[report.location, report.lake, report.category]],
10  });
11
12  autoTable(doc, {
13    startY: 50,
14    head: [["Disease Name", "Description"]],
15    body: report.diseases.map((d) => [d.name, d.description]),
16  });
17
18  doc.save("Health_Report.pdf");
19 }

```

Listing 7.4: Generate and Download Health Report PDF

## 7.3 Implementation Screenshots

### 7.3.1 Landing Page

The WBPA-QA application starts on the main page, which offers users an easy interface to learn about water quality monitoring and access pertinent functions.

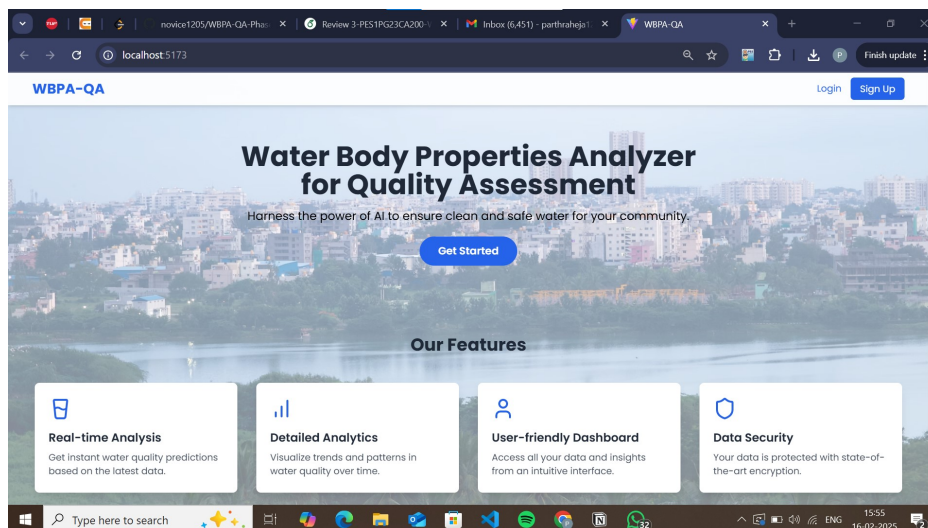


Figure 7.1: Landing Page of WBPA-QA

### 7.3.2 User Dashboard

Users of the dashboard can access several capabilities including water quality metrics, health predictions, and suggestions by navigating to different points of view of the system.

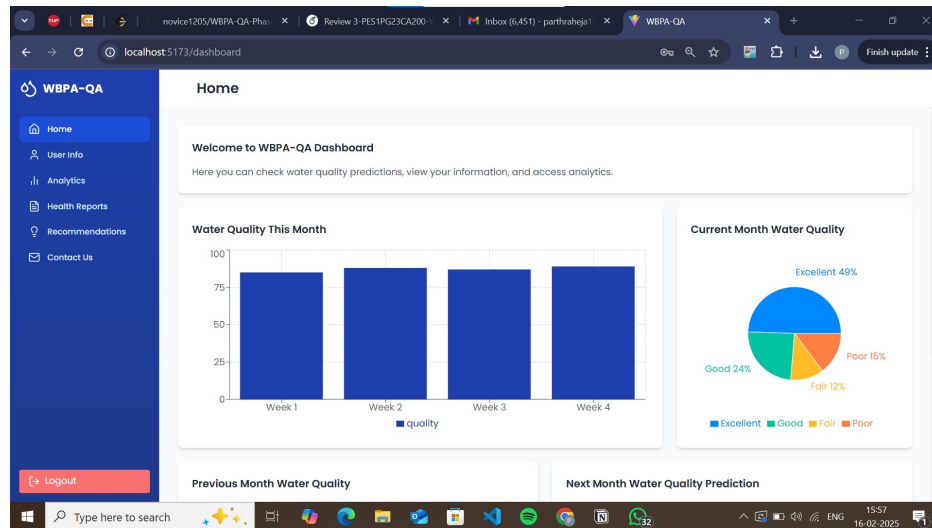


Figure 7.2: User Dashboard with Navigation Features

### 7.3.3 Analytics with Health Predictions

Along with health-related insights obtained from expected water quality indicators, this part shows water quality analytics depending on user input and trained ML models.

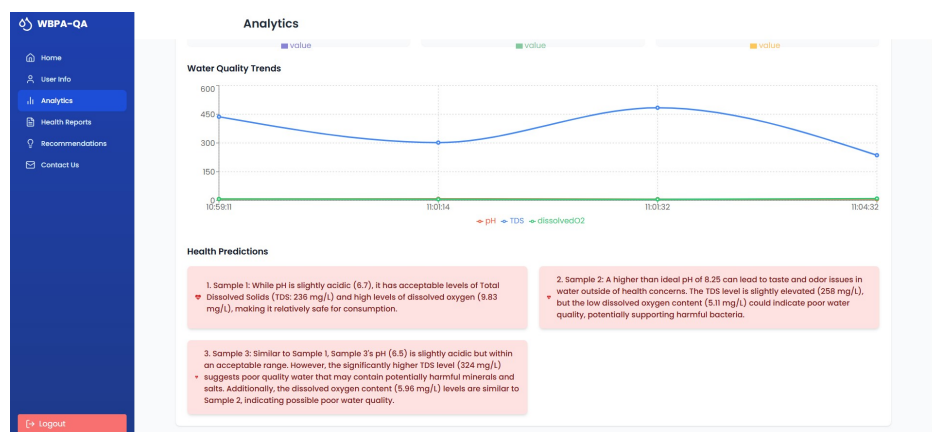


Figure 7.3: Analytics Page with Water Quality Predictions

### 7.3.4 Health Reports Generation and Download

Depending on the expected water quality data for their chosen area, users can create and download customised health reports.

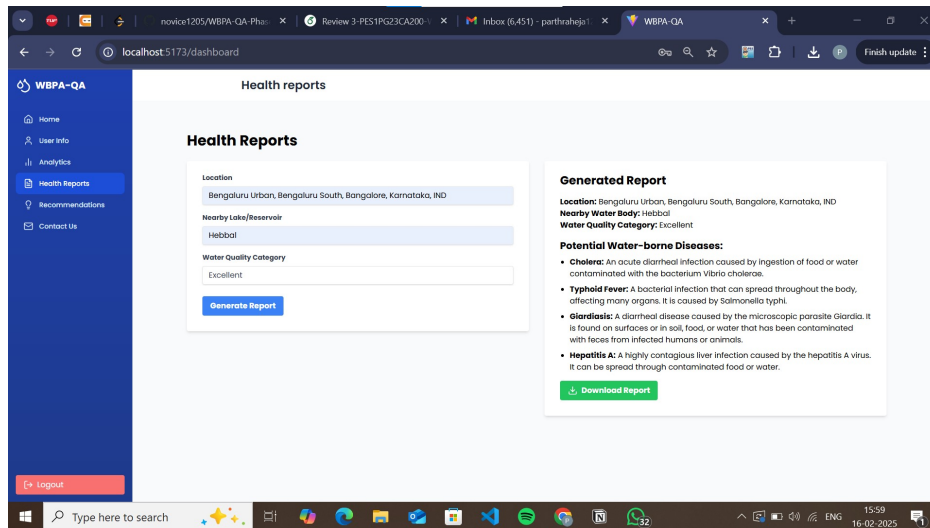


Figure 7.4: Health Reports Generation and Download Page

### 7.3.5 Recommendations Page

Users get tailored advice including suggested filtration techniques, dietary adjustments, and preventative actions based on water quality results.

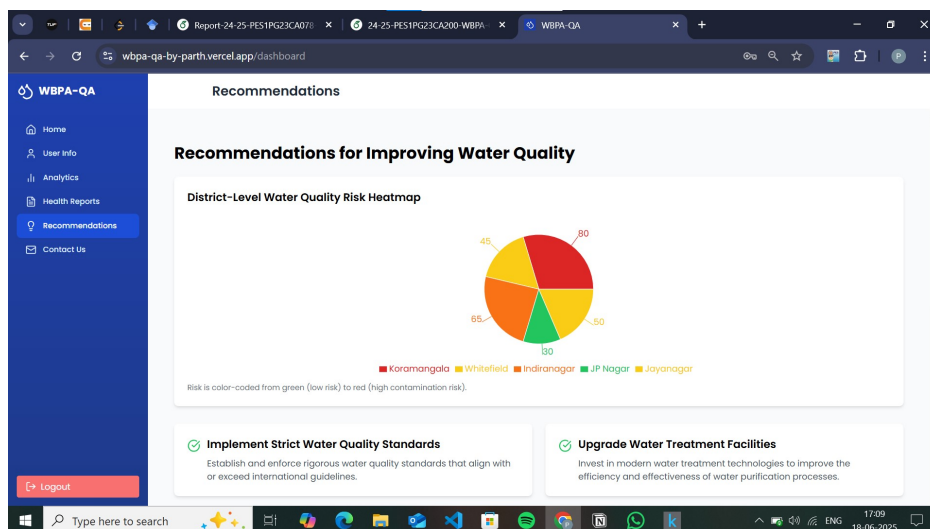


Figure 7.5: Recommendations Page with Health and Safety Advice

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WBPA-QA

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## Chapter 8: Testing

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# Chapter 8

## Testing

### 8.1 Manual Testing

#### Test Case: Water Quality Analysis and Risk Advisory Display

This test checks that the system appropriately calls the ML model, gets the forecast, shows graphs, and creates suitable health advice when valid data is entered into the analytics form.

| MANUAL TESTING                          |                                                   |                                                                    |                                             |                                      |        |
|-----------------------------------------|---------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------|--------------------------------------|--------|
| MODULE: Analytics and Health Prediction |                                                   |                                                                    |                                             |                                      |        |
| Test ID                                 |                                                   | TC-M01                                                             |                                             |                                      |        |
| Test Name                               |                                                   | Submit Parameters and Check Prediction Flow                        |                                             |                                      |        |
| Description                             |                                                   | Ensure valid predictions are shown when correct data is submitted. |                                             |                                      |        |
| #                                       | Step                                              | Data                                                               | Expected Output                             | Actual Output                        | Status |
| 1                                       | Open Analytics Page                               | –                                                                  | Analytics page should load without error    | Page loads correctly                 | Pass   |
| 2                                       | Submit valid values (pH, TDS, DO, BOD, Turbidity) | pH: 6.8, TDS: 500, DO: 6, BOD: 3, Turbidity: 4                     | Inputs accepted; no validation error        | Inputs validated successfully        | Pass   |
| 3                                       | Click "Predict"                                   | –                                                                  | Model returns water class and risk advisory | Quality and health message displayed | Pass   |
| 4                                       | View graphs                                       | –                                                                  | Graphs show parameter ranges with labels    | Graphs are correct and responsive    | Pass   |
| 5                                       | Submit empty form                                 | –                                                                  | Should show validation message              | Error not displayed clearly          | Fail   |
| 6                                       | Input out-of-range pH                             | pH = 25                                                            | Show warning or block submission            | Input accepted without warning       | Fail   |

Table 8.1: Test Case: Analytics Functionality

# Test Case: Graph Responsiveness on Varying Screen Sizes

This test confirms whether graphs of water quality remain responsive throughout a range of screen sizes.

| MANUAL TESTING                  |                           |                                                                         |                                           |                          |        |  |
|---------------------------------|---------------------------|-------------------------------------------------------------------------|-------------------------------------------|--------------------------|--------|--|
| MODULE: Dashboard Visualization |                           |                                                                         |                                           |                          |        |  |
| Test ID                         |                           | TC-M02                                                                  |                                           |                          |        |  |
| Test Name                       |                           | Graph Rendering on Mobile Devices                                       |                                           |                          |        |  |
| Description                     |                           | Ensure the graphs are properly scaled and do not overflow screen width. |                                           |                          |        |  |
| #                               | Step                      | Data                                                                    | Expected Output                           | Actual Output            | Status |  |
| 1                               | Open dashboard on desktop | –                                                                       | Graphs fully visible                      | Fully visible            | Pass   |  |
| 2                               | Open dashboard on mobile  | –                                                                       | Graphs should shrink/rescale responsively | Slight overflow observed | Fail   |  |

Table 8.2: Test Case: Responsive Visualization



## 8.2 Automated Testing

### Test Case: PDF Health Report Generation

This test confirms that the system effectively creates a downloadable PDF report with all the necessary elements in a structured way following a forecast.

| AUTOMATED TESTING         |                                 |                                                                                   |                                              |                                         |             |
|---------------------------|---------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------|-------------|
| MODULE: Report Generation |                                 |                                                                                   |                                              |                                         |             |
| Test ID                   |                                 | TC-A01                                                                            |                                              |                                         |             |
| Test Name                 |                                 | Verify PDF Generation after Prediction                                            |                                              |                                         |             |
| Description               |                                 | Ensure the report is generated with prediction and advisory in proper PDF format. |                                              |                                         |             |
| #                         | Step                            | Data                                                                              | Expected Output                              | Actual Output                           | Status      |
| 1                         | Complete a valid prediction     | Sample input (pH, DO, etc.)                                                       | Receives quality + advisory                  | Output classification + advisory shown  | Pass        |
| 2                         | Click “Download Report”         | –                                                                                 | PDF download initiated with layout           | Report download triggered               | Pass        |
| 3                         | Open PDF                        | –                                                                                 | File has inputs, prediction, advisory layout | Layout is readable and correct          | Pass        |
| 4                         | Check filename                  | –                                                                                 | File is named properly                       | File name is correct (e.g., report.pdf) | Pass        |
| 5                         | Test with blank prediction data | –                                                                                 | Should not allow PDF export                  | No validation message shown             | <b>Fail</b> |

Table 8.3: Test Case: PDF Report Generation

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WBPA-QA

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## Chapter 9: Conclusion

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## Chapter 9

# Conclusion

Particularly in fast urbanising areas like Bengaluru, water quality is a vital sign of environmental sustainability and public health. Designed to close the distance between intelligent, AI-driven analytics and hand water monitoring, the WBPA-QA (Water Body Property Analyser for Quality Assessment) system This research links the prediction with real-time, user-friendly visualisations and advise creation using NLP and effectively combines machine learning models for classifying water quality based on fundamental criteria like pH, TDS, DO, BOD, and turbidity.

The system was launched as a full-stack online application with a neat, responsive interface catering municipal officials as well as the general public. Users can get organised reports, examine water samples, get thorough forecasts and health recommendations. The project underwent thorough manual and automated testing; Random Forest obtained a 91% accuracy rate, therefore proving both technical resilience and practical relevance.

WBPA-QA shows generally that artificial intelligence-based technologies can turn slow, scattered environmental monitoring from passive, data-driven decision-making system into proactive one. It could help local government, raise community awareness, and fit with national sustainability objectives.

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WBPA-QA

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## Chapter 10: Future Work

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# Chapter 10

## Future Work

Although WBPA-QA shows encouraging outcomes, there are still various chances to improve the performance, reach, and influence of the system going forward.

Integration of IoT-based water sensors that can send real-time parameter data from several sites straight into the system is one main area for expansion. This would eliminate the reliance on hand data entry and thereby increase early contamination alert reaction times. Support for regional languages such Kannada, Hindi, and Tamil will also help non-English speaking populations become more accessible and adoptable.

The creation of a mobile application with location-aware capabilities that can automatically retrieve surrounding lake data and alert consumers about water safety offers even another interesting path. Moreover, incorporating satellite data for spatial analysis and extending prediction categories for more particular diseases would offer closer knowledge on long-term environmental and public health trends.

At last, cooperation with local municipal authorities and integration into government dashboards—such as Smart City platforms—would boost the scope and practical value of WBPA-QA, so establishing WBPA-QA as a fundamental tool for sustainable urban development.

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WBPA-QA

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## Appendix A: Bibliography

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# Appendix A: Bibliography

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WBPA-QA

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## Appendix B: User Manual

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# Appendix B : User Manual

## Overview

The WBPA-QA application is a smart water quality monitoring and risk prediction system designed to assist both public users and authorities. It provides insights into water usability, forecasts key parameters like pH and TDS, and generates personalized health recommendations.

This manual provides step-by-step guidance for using the system effectively. **System**

## Requirements

- **Operating System:** Windows 10+, macOS, Linux (Ubuntu 18.04+)
- **Web Browser:** Chrome, Firefox, Edge (latest version)
- **Backend:** Python 3.8+, Node.js 18+, MongoDB Atlas, Flask API
- **Frontend:** React.js (with Tailwind CSS), Axios, Chart.js
- **Hardware:** Minimum 4GB RAM, 2GHz processor

## Installation Instructions

### For Developers (Local Setup)

1. Clone the repository from GitHub: `git clone https://github.com/your-repo/wbpa-qa.git`
2. Install backend dependencies: `cd backend && pip install -r requirements.txt`
3. Install frontend dependencies: `cd frontend && npm install`
4. Run the backend server: `python app.py`
5. Run the frontend: `npm run dev`
6. Access the app at `http://localhost:3000`

### For End Users (Public Portal)

No installation is required. Access the live deployment at:  
`https://wbpa-qa-by-parth.vercel.app/`

# How to Use the Application

## 1. Public Users

- Open the web portal and navigate to “Check Water Quality”.
- Enter your STN Code or location and select a month.
- View predicted parameters: pH, TDS, Use Class.
- Access health suggestions and recommended filters.
- Report any observed pollution through the “Raise Complaint” section.

## 2. Authorities

- Log in using credentials provided by the administrator.
- View real-time dashboards, pollution hotspots, and past records.
- Generate reports and download analytics.
- Manage public complaints and monitor resolution status.

## Troubleshooting

- **Issue:** “Unable to connect to server” *Solution:* Ensure backend server is running and internet is stable.
- **Issue:** “Predictions not loading” *Solution:* Check if the model is deployed and MongoDB is connected.
- **Issue:** “Login Failed” *Solution:* Check email/password. Contact admin for credential reset.

## Support

For support or queries, contact: **Email:** parthraheja1205@gmail.com **Phone:** +91-8826202212  
**Website:** <https://wbpa-qa-by-parth.vercel.app/>

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*Note: This user manual is subject to updates with future application versions.*

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WBPA-QA

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# Appendix C: Plagiarism and AI Report

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# Appendix C: Plagiarism and AI Report

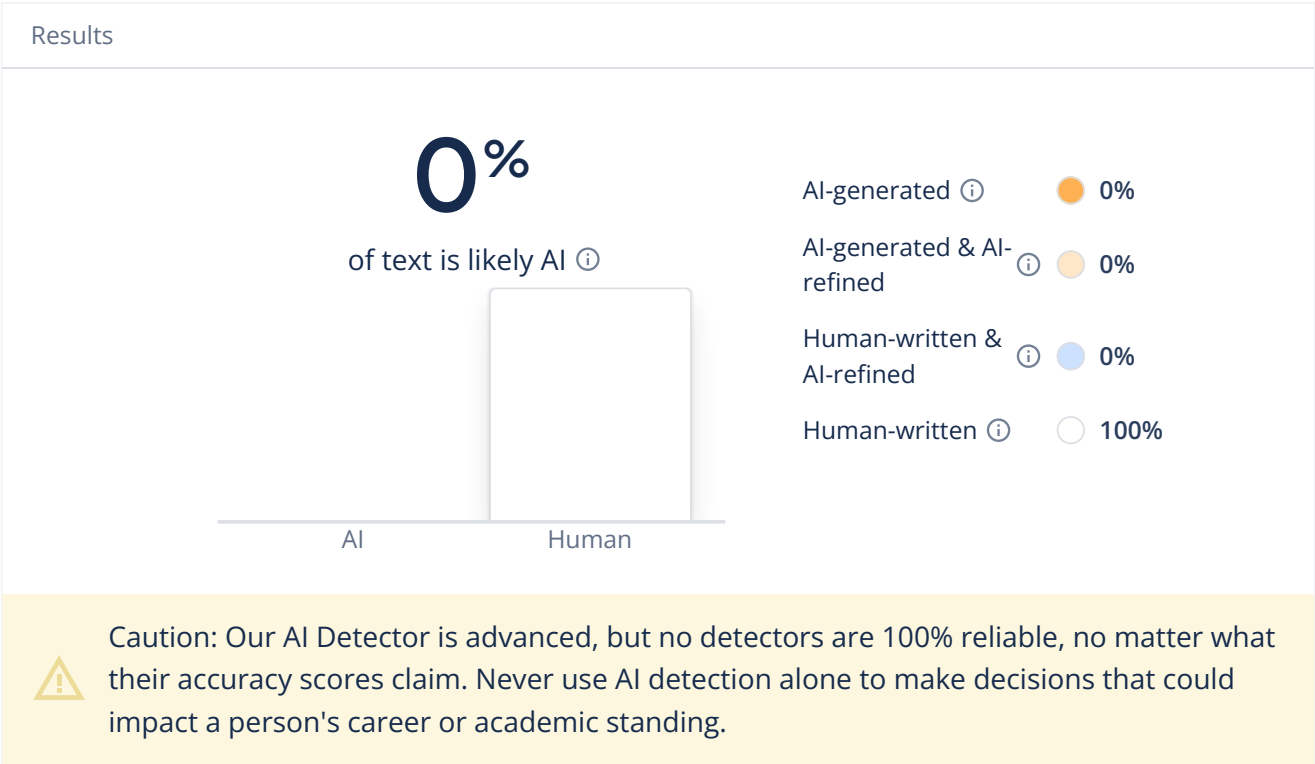
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## Appendix D: Poster

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# Appendix D: Poster

## PES UNIVERSITY DEPARTMENT OF COMPUTER APPLICATIONS



# WBPA-QA

## WATER BODY PROPERTY ANALYZER FOR QUALITY ASSESSMENT

**BY - PARTH RAHEJA**  
MCA, SRN - PES1PG23CA200

**GUIDE - MR. DILIP KUMAR MARIPURI**  
Associate Professor, MCA

### ABSTRACT

WBPA-QA is a smart, web-based water quality monitoring and health risk prediction platform that uses machine learning to assess key parameters like pH, TDS, and Dissolved Oxygen. The system offers real-time analytics, predicts potential waterborne diseases, and generates health reports.



### METHODOLOGY

- User roles include general public and administrative authorities
- Location and lake-specific selection enables contextual analysis
- Predictive ML models forecast water quality index and disease risks

### TOOLS & TECHNOLOGIES



### CONCLUSION

WBPA-QA bridges the gap between environmental monitoring and public health using AI-powered predictions and data visualization. It provides a scalable, accessible solution for water quality awareness and enabling smarter water resource management.